DRUGS · SPECIAL ENT AGENTS

Heliox

Heliox is a mixture of helium and oxygen (commonly 79:21, 70:30 or 60:40). Helium is far less dense than nitrogen, so the mixture flows through a narrowed airway with less turbulence and less work of breathing. In upper airway obstruction it can buy time while the cause is treated or a definitive airway is planned. It treats nothing itself, and it limits the oxygen that can be given.

1 CHAPTER ONE At a glance

The gas on one page.

1.1 Summary

HELIOX

	SUMMARY
Composition	Helium with 21% to 40% oxygen
How it works	Low density reduces the Reynolds number, converting turbulent to more laminar flow through a stenosis; resistance and work of breathing fall
ENT uses	Temporising severe upper airway obstruction (tumours, angioedema, subglottic stenosis, post-extubation stridor, croup) while preparing definitive treatment; supporting breathing during transfer
Delivery	Tight-fitting non-rebreathing mask at high flow; flowmeters calibrated for heliox
Limitations	The more oxygen needed, the less helium can be given (benefit falls when oxygen requirement exceeds about 40%)

2 CHAPTER TWO Using it in ENT

Practice.

2.1 In the airway emergency

Heliox improves stridor and distress, sometimes dramatically, but it can mask how bad the obstruction is and lull the team. Use the time to bring the right people (senior ENT surgeon, anaesthetist), plan the airway (awake fibreoptic intubation, awake tracheostomy, or front-of-neck access), and treat the cause (steroids, nebulised adrenaline, antibiotics, drainage).

3 CHAPTER THREE Safety

What goes wrong.

3.1 Problems

Hypoxia if the oxygen fraction is too low for the patient's needs; a falsely reassuring clinical picture; voice distortion (high pitch).

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

> WHAT THE EXAMINER IS LOOKING FOR
>
> - The physics (density and turbulent flow).
> - That heliox is a **bridge**, not a treatment.

4.2 Common pitfalls

COMMON PITFALLS

- Using heliox instead of securing the airway.

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